Zhao Zehui, Leila

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Mechanical Engineer experiences in mechanical design and rapid prototyping. Passionate in Product development in Space Systems, Robotics, UAV Design, CFD, MDO.

EDUCATION

The University of Hong Kong, Bachelor's in Mechanical Engineering 2022 – now

- Highest Attained Semester GPA: 3.36/4.3
- Scholarship: Tam Wing Fan Innovation Academy Funding 2025 – 2026: 100K HKD
- Course taken: Mechanics of Solid (A), Engineering Design and Manufacture (A+), Unmanned Aerial Vehicle design (A+), Dynamics and Control (A-), Thermodynamics (A-), Aircraft Stress Analysis and Finite Element (master course) (A)

WORK & RESEARCH EXPERIENCES

The Hong Kong Productivity Council **Hong Kong**
Winter Intern in Smart Manufacturing Division *Dec 2025 – Jan 2026*

- Assisted in an ESG and textile recycling toolkit and developed survey by analyzing global regulations, supporting circular innovation for Hong Kong fashion SMEs and AI integration.
- Designed a 3D camera plus multi-axis robotic arm mounting system using SolidWorks (2D/3D drawings, BOM and GD&T)
- Developed applied AI for industrial thermographic anomaly inspection on 6-axes robotic arms, with an accuracy of .
- Designing a conceptual SLAM-based indoor drone inspection project proposal based on mechanical data (FEA) to detect internal building cracks.

Faculty of Engineering, the University of Hong Kong **Hong Kong**
Part-time Temporary Student Research Assistant *Sept 2025 – Apr 2026*

- Mentored over 30 students in robotic arm control and design, FDM 3D printing and CAD software utilization, enhancing their practical skills and understanding of engineering concepts.
- Provide constructive feedback to lecturers on behalf of workshop content which assisted them in workshop development.

Feelings Group Limited | Hong Kong Science & Technology Park **Hong Kong**
Student Technician – Hardware team *Dec 2024 – Jan 2025*

- Provided academic and industrial research for hardware selection and market database of radar system for medical devices tracking vital signs of small animals.
- Assisted in CCTV development for the medical device, using electronic modules, such as Raspberry Pi, LAN network and camera module.

WeGoAlgae **Hong Kong**
Mechanical and Electrical Engineer *Jul 2024 – Jan 2025*

- Designed a conceptual cylinder water tank with water-proof turbine for improving water circulation rate and algae photosynthesis efficiency.
- Provided constructive consultation to the mechanical and electrical design of the algae development container, familiar with mini pumps (5V) simple circuit design; monitor the algae's growth and the environment.

The Hong Kong and China Gas Company Limited (Towngas) **Hong Kong**
Summer intern in Residential Product & Quality department *Jul - Aug 2024*

- Contributed to the development of 7 IoT gas appliances, and 10+ gateways.
- Identified potential problems in functional tests and provided effective suggestions to enhance user-friendliness of the mobile app interface.
- Assisted in the assessment of gas appliance accessories such as flexible hose and gas malleable iron fittings.
- Analyzed market potential for company's products and participated in daily meetings, contributed to strategic planning and understanding team structure and working principles and provided valuable feedback to supervisor regarding the market demand and daily workflow.

RESEARCH EXPERIENCES & MANUSCRIPTS

Multidisciplinary Design Optimization Framework for the AIAA Design-Build-Fly 2026 Competition **Hong Kong**
First Author (Submitted on 21st May 2026 to AIAA SciTech Forum 2027) *Nov 2025 - now*

- A parametric Multidisciplinary Optimization (MDO) framework was developed in MATLAB using a deterministic grid-search approach to explore the design space systematically to obtain the predicted performance of the preliminary design that maximizes mission score while satisfying aerodynamic and performance constraints.

MAJOR AEROSPACE PROJECTS & COMPETITIONS

HKU Design, Build and Fly Team **Hong Kong, USA**
HKU AIAA DBF Competition Team Leader *Aug 2025 – May 2026*

- Led **aerodynamic** analysis and structural wing design under the constraint of 1.25 and 1.5-meter wingspan for an 8kg and 2.7kg aircraft and fast-speed flight.
- Conducted airfoil, **CFD** and stability analysis using XFLR5 and Ansys to improve flight performance.
- Designed banner mechanisms to stow, deploy, tow and release 4-m long banner in flight utilizing SW, **FDM 3D Printing** and Arduino.
- Built a MATLAB-based multidisciplinary optimization framework to automate mission score optimization with assistant of AI tools.
- Ranked **32nd out of 98** teams from universities worldwide competing in AIAA DBF 2026, **3rd** in Asia and **2nd** in Hong Kong.

HKU DBF Student Interest Group Leader *Sept 2024 – now*

- Conducted over 30 workshops and training to promote the team to new members, and received 100K HKD funding from Innovation Academy, HKU.
- Secured three industry sponsorships, including Ansys, L'Voyage and the Safety Collaborative for DBF2025-2026.
- Manage the team's human resources, documents, promotion and train next year's leaders to enlarge the team size and development.
- Conducted research study on 5-8 3D printing material regarding the topic of lightweight design and analysis for ribs construction in fixed-wing model aircraft, tested on **FDM 3D printer (Bambu Lab X1C & H2D**).
- Designed team jacket and posters for DBF 2025 & 2026.
- Manage financial documents and budget planning 2025 – 2026.
- Advocated 4 new aircraft design competition team and student chapter for team members to enhance their professional development and connections.

MATLB Interstellar Project **Dec 2024**

- Analyze stellar motion by data plotting regarding intensity and wavelength of 7 stars to calculate their doppler effect, and compare their stellar spectra provided in the database.

RAICOM Robotic Arm Challenge (Shenzhen) **Jul 2025**

- Broke down the structure of the algorithm for automated and semi-automated missions; performed data annotation for AI model training.

CERTIFICATES & AFFILIATIONS

Royal Aeronautical Society (HK) Student Member 2025
IMechE Affiliated Member & Committee 2025
HKIE Aircraft Division – Student Member 2025 – 2026
General Secretary, Publication and Publicity Secretary, Physics Society, HKU (2023)
Peking University Summer Exchange 2023
HKU Lead-For-Life Cohort Member 2022 – 2026

SKILLS

Language: Chinese (Mandarin, Cantonese): Native, English: Proficient, German: Basic
CAD Software: SolidWorks (proficient), AutoCAD (proficient), fusion360 (intermediate)
CFD Software: Xflr5 (proficient), Ansys Fluent (proficient)
Other Software: HIKIMICRO Studio (intermediate), Bambu lab Studio (proficient), Microsoft Office Suite (proficient), Roboflow (intermediate)
Programming: MATLAB (intermediate), Arduino (intermediate), Python (intermediate), Orange data mining (beginner), Visual studio (intermediate)